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SFAD: Speculative Factuality-Aware Decoding

A DPO-Trained Draft Model Plus Epistemic Friction Enforces Contextual Faithfulness at Speculative Decoding Speeds

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Contextual faithfulness — the property that a language model’s output is consistent with a provided source document — is critical for retrieval-augmented generation, summarisation, and question answering. Existing mitigation strategies either double inference cost (contrastive decoding) or require expensive post-training (RL alignment). SFAD achieves a 38% relative reduction in hallucination rate and an 11.6-point FactScore gain over greedy decoding, at throughput matching standard speculative decoding, by training a factuality-specialised draft model with fine-grained preference data and enforcing faithfulness online via an Epistemic Friction detector.

AT A GLANCE

Problem: LLMs hallucinate even when source documents are provided; existing remedies sacrifice inference speed.

Why it matters: RAG and summarisation systems used in production cannot tolerate either hallucinations or latency overhead.

Method: DPO-trained context-faithful draft model (ConFide data) + Epistemic Friction detector; no change to verifier LLM or workflow.

Result: FactScore 72.8% vs. 61.2% greedy; 88 tok/s vs. 91 tok/s speculative decoding (no throughput penalty).

Evidence: RAGTruth, TruthfulQA, SummEval benchmarks; ablations isolating ConFide DPO and Epistemic Friction contributions.

The Big Picture

Retrieval-augmented generation (RAG) grounds language model outputs in retrieved source documents. Despite the presence of relevant documents in context, large language models still hallucinate — generating plausible-sounding but contextually unfaithful text that contradicts or is unsupported by the source.

The challenge is threefold. First, hallucination occurs not because the model lacks information but because the

decoding process does not explicitly enforce consistency with the context. Second, the most effective existing methods (contrastive decoding) require two forward passes, doubling inference cost. Third, post-training alignment via RL requires significant compute and careful reward design.

SFAD is the first approach to embed factuality enforcement inside speculative decoding, achieving faithfulness gains with zero throughput overhead.

Why Existing Approaches Fall Short

Contrastive decoding compares model outputs under different prompts (faithful vs. non-faithful) and selects tokens with higher probability under the faithful prompt. It requires two forward passes per token, doubling inference cost — impractical in production.

RL post-training alignment trains the model to prefer faithful outputs via reward models, but requires thousands of feedback examples and careful reward shaping. The resulting model still hallucinate under distribution shift.

Standard speculative decoding uses a small draft model to propose multiple tokens for a large verifier LLM to accept/reject in parallel. This achieves 2–3× speedup but makes no claim about the faithfulness of accepted tokens.

Core Mechanism

SFAD modifies the draft model in speculative decoding to be *context-faithful*, so that the token proposals it generates are more likely to be grounded in the source document. The verifier LLM then accepts these proposals with high probability — because they are both fluent (the draft is a capable LLM) and faithful (the draft is DPO-trained on faithfulness preferences). No modification to the verifier or the acceptance criterion is required.

An additional online component, Epistemic Friction, detects tokens where the draft’s proposal diverges from the verifier’s preference in a context-relevant direction, and steers the draft before it proposes — further increasing acceptance rate and faithfulness.

Technical Formulation

4.1 ConFide Dataset

ConFide is a preference dataset of (context, chosen output, rejected output) triples, constructed from model outputs on knowledge-intensive prompts via *atomic perturbations*:

- **Entity swap:** replace entity names with plausible alternatives from the knowledge base.
- **Relation inversion:** change predicate direction (e.g. “A was founded by B” → “B was founded by A”).

- **Quantifier change:** replace “all” with “some”, exact numbers with nearby values.
- **Temporal shift:** shift dates or tenses.

Original outputs (faithful to source) are chosen; perturbed outputs are rejected. This produces fine-grained atomic contrasts that teach the draft model exactly which token sequences are unfaithful.

4.2 DPO Training

The draft model π_θ is fine-tuned on ConFide with Direct Preference Optimisation:

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E}_{(x, y_+, y_-)} \left[\log \sigma \left(\beta (\log \pi_\theta(y_+|x) - \log \pi_{\text{ref}}(y_+|x)) - \beta (\log \pi_\theta(y_-|x) - \log \pi_{\text{ref}}(y_-|x)) \right) \right] \quad (1)$$

where y_+ is the faithful (chosen) output, y_- is the perturbed (rejected) output, and π_{ref} is the pre-DPO draft.

4.3 Epistemic Friction

At each proposed token y_t , the Epistemic Friction score is:

$$\text{EF}(t) = \text{KL}(\pi_{\text{draft}}(\cdot|x, y_{<t}) \parallel \pi_{\text{verifier}}(\cdot|x, y_{<t})) \cdot w_{\text{specialist}}(t)$$

where $w_{\text{specialist}}(t)$ is the verifier’s mean attention weight to the source document at position t , measuring how context-relevant the current generation position is.

High $\text{EF}(t)$ flags a likely hallucination site. The draft’s sampling temperature is increased at these positions to diversify proposals away from the unfaithful mode.

4.4 Speculative Acceptance

The standard speculative decoding acceptance criterion is unchanged:

$$P(\text{accept } y_t) = \min \left(1, \frac{\pi_{\text{verifier}}(y_t|x, y_{<t})}{\pi_{\text{draft}}(y_t|x, y_{<t})} \right)$$

Because the DPO draft assigns higher probability to faithful tokens, the verifier’s acceptance rate increases for faithful proposals and decreases for unfaithful ones — without any change to the acceptance formula.

Learning or Inference Procedure

Training:

1. Sample model outputs on knowledge-intensive prompts.
2. Apply atomic perturbations to construct ConFide preference pairs.
3. Fine-tune a small draft LLM (3–7B parameters) with DPO on ConFide.

Inference:

1. Draft model proposes $k = 4$ tokens per speculative step.
2. Compute EF for each proposed token; re-sample tokens above the EF threshold.
3. Verifier LLM processes the k (steered) tokens in parallel and accepts/rejects via the standard criterion.
4. Accepted tokens are output; rejected tokens restart from the last accepted position.

The EF computation adds a single matrix product over the verifier’s cached attention weights — negligible overhead.

What the Guarantee Says

The paper proves that under DPO training, the faithful token acceptance rate improves by at least:

$$\Delta_{\text{accept}} \geq \frac{\beta}{Z} (\mathbb{E}[\log \pi_{\text{DPO}}(y_+)] - \mathbb{E}[\log \pi_{\text{ref}}(y_+)])$$

which is strictly positive whenever DPO training increases the faithful log-probability above the reference model — a necessary condition for successful DPO training. The bound is tight for token-level independent acceptance.

Experimental Findings

Main results across benchmarks:

Method	FactScore↑	Throughput
Greedy (verifier only)	61.2%	38 tok/s
Contrastive decoding	68.4%	19 tok/s
Standard speculative decoding	61.3%	91 tok/s
SFAD (ours)	72.8%	88 tok/s

SFAD improves FactScore by 11.6 points over greedy and 4.4 points over contrastive decoding, while matching standard speculative decoding throughput. On RAGTruth specifically, hallucination rate drops from 28.3% to 17.5% (38% relative reduction).

Ablations and Interpretation

Without ConFide DPO: FactScore drops by 6.8 points — the draft model fine-tuning is the primary source of faithfulness gain.

Without Epistemic Friction: FactScore drops by 2.1 points, showing that online steering provides a complementary, additive faithfulness layer.

Atomic vs. coarse perturbations in ConFide: Fine-grained atomic perturbations outperform sentence-level contrastive pairs by 3.4 FactScore points, validating that granular contrast is essential for teaching the draft model about specific faithfulness failures.

Specialist weight $w_{\text{specialist}}$: Attention-based specialist

weighting outperforms uniform weighting by 1.8 FactScore points. The model is most useful at context-relevant positions, not uniformly.

EF threshold sensitivity: FactScore is robust to the EF threshold in the range $[0.3, 0.7]$; below 0.3 too many tokens are resampled (hurting fluency); above 0.7 too few hallucinations are caught.

Reference

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